



1. Executive Context & Mission Statement:

Traditional tech hub internship cohorts experience significant communication and task fragmentation when relying on informal messaging channels (such as WhatsApp or Telegram groups) and scattered cloud folders. Lost assignments, missed lecture schedules, zero role boundary enforcement, and cross-cohort noise create substantial operational friction.

Knowvia solves this by consolidating core academic workflows into an isolated, department-scoped Progressive Web Application (PWA). Every department (Frontend, Backend, Cybersecurity, UI/UX, and Data Science) operates in complete workspace isolation, governed by fine-grained Role-Based Access Control (RBAC), real-time WebSocket communication, verified file sharing, and automated notification services.

5 / 5

ISOLATED WORKSPACES

42 / 42

TESTS PASSING (100%)

100% Free

FREE-TIER CLOUD INFRA

0 Demo Data

CLEAN DB FOOTPRINT

1 SYSTEM ARCHITECTURE & CORE TECHNOLOGY STACK

Layer / Subsystem	Technology Stack	Architectural Role & Scope	Deployment & Cost Profile
Client Interface	React 19, TypeScript, Vite, React Router v7	Role-based views (Admin, Tutor, Intern), vanilla CSS design tokens, zero external UI kit bloat.	Vercel Free Tier (Global Edge CDN, SSL)
Backend API	Node.js, Express.js, TypeScript, Zod	REST API endpoints, JWT token authentication, input sanitization, error-handling middleware.	Render Free Tier (Automated Git deploys)
Real-Time Messaging	Socket.io (WebSockets)	Department-scoped rooms (dept:{slug}), personal alerts (user:{id}), typing state.	Integrated into Express service instance
Data Persistence	Prisma ORM, Supabase PostgreSQL	Relational schemas for users, departments, assignments, schedules, submissions, and reviews.	Supabase Free Tier (Managed PostgreSQL)
File Screening	Multer (Memory), Custom Magic-Byte Validator	Binary signature inspection, executable blocklist (.exe, .sh), UUID storage naming.	Supabase Storage S3-Compatible / Local
Push & Schedules	Web Push (VAPID), Node-Cron	Hourly background cron daemon delivering 24-hour class reminders and event notifications.	In-process daemon with automatic retry

```
[Knowvia Bootstrap] Server listening on port 4000 (NODE_ENV=production)
🔑 Administrator Verified: Name: NASCOM | Email: admin@knowvia.internal | Role: ADMIN
[Socket.io] Real-time gateway active across 5 department channels.
[Node-Cron] Class reminder background daemon initialized (interval: hourly).
```

2 COMPLETED FUNCTIONAL MODULES & ENGINEERING MILESTONES

1. Admin-Controlled Email Onboarding

- Eliminated vulnerable public self-registration.
- Admin generates SHA-256 encrypted single-use invitation tokens.
- Automated invitation dispatch via Nodemailer SMTP service.
- Secure activation page (/onboarding?token=...) allowing users to establish their identity and password securely.
- Comprehensive User Directory with soft deletion (deletedAt, isActive = false) and self-deletion safeguards.

2. Assignment Management & Review Flow

- Replaced cumbersome project boards with a streamlined academic assignment workflow.
- Tutors create tasks with instructions, due dates, and 25MB file limits.
- Interns submit work with attached files and commentary.
- Multi-state evaluation cycle: SUBMITTED → IN_REVIEW → APPROVED / NEEDS_REVISION / REJECTED.
- Interactive student progress meter on intern dashboards.

3. Class Scheduling & 24h Cron Reminders

- Interactive scheduling engine for physical hub sessions and virtual video meetings.
- Dedicated node-cron background service executing hourly.
- Identifies classes starting within the next 24 hours and dispatches automated reminder notifications to enrolled department interns.
- Status tracking prevents duplicate reminder dispatch.

4. Enterprise File Screening Engine

- Accepts diverse educational formats while guaranteeing server safety.
- Binary magic-byte inspection confirms genuine file headers against declared MIME types (blocks masked malware).
- Strict extension blacklist rejects .exe, .bat, .cmd, .sh, .ps1, .dll, etc.
- Path-traversal sanitization and UUID file isolation.

5. Real-Time Chat & Live Interaction

- Department-scoped Socket.io chat rooms with instant message delivery.
- Live typing state indicators (user:typing) and reply threading.
- Dual-mode message delivery with persistent PostgreSQL database storage for historical retrieval and seamless reconnection handling.

6. Deep-Linked Announcement Engine

- Multi-tier announcements: Global broadcasts (Admin/NASCOM) and department-specific notices (Tutors).
- Automated system announcements generated upon assignment publishing, class scheduling, or material uploading.
- Smart deep linking: Clicking notifications automatically routes and scrolls the viewport directly to the referenced content.

3 IDENTITY GOVERNANCE & UI/UX HARDENING

Elimination of Demo UI & Canonical Administrator Identity:

All mock login selectors, demo account buttons, and role-switching shortcuts were permanently decommissioned. The primary administrator was updated to the canonical organization entity **NASCOM**. Credentials are never rendered on the frontend or exposed in JavaScript bundles.

System-Wide Initials-Only Avatar Engine

- Profile photo uploads completely deprecated to eliminate unnecessary cloud storage fees and safeguard intern privacy.
- Implemented synchronized canonical avatar engines across client and server:
 - Single name: "NASCOM" → "N"
 - Full name: "John Doe" → "JD"
 - Multi-word: "Mary Jane Williams" → "MW"
 - Null/Whitespace fallback: "U" (prevents "undefined").
- Consistent, role-tinted circular badges across top navigation, chat channels, and directories.

Permanent Elimination of Mock Data

- Traced and resolved recurring test-user resurrection in automated test fixtures.
- Completely purged mock users (David Kim, Marcus Chen, Maya Patel, Jordan Lee, Alex Vance) from seed scripts, production migrations, and test environments.
- Test suites refactored with ephemeral isolation: automated tests generate transient records that are completely wiped upon completion.

4 PWA INSTALLABILITY, WEB PUSH & DEVICE SECURITY

Progressive Web App (PWA) Standards

- Validated Web App Manifest (manifest.json) with standalone display mode, theme colors, and education/productivity category tags.
- Service worker (sw.js) configured with asset pre-caching, fetch event listener handling, and offline fallbacks.
- Cross-platform support: iOS Safari meta declarations and desktop/Android install prompt lifecycle management.

Web Push & Multi-User Device Safeguards

- Background Web Push via VAPID keys reaches devices even when the PWA or browser is closed.
- Device Dissociation on Logout: Prevents notification leakage when multiple users share a single hub workstation. Subscriptions are safely dissociated on logout and re-linked upon re-authentication without token corruption.

5 VERIFICATION, BUILD HEALTH & QA METRICS

Validation Category	Verification Command / Tool	Result Status	Operational Impact
Avatar Engine Tests	npm --prefix server test	18 / 18 PASSED	Verified initials formatting, multi-word strings, null guards, and clean display rendering.
Auth & Onboarding Tests	npm --prefix server test	24 / 24 PASSED	Verified SHA-256 token hashing, email activation, soft-deletion, and self-removal guards.
Frontend Build	npm --prefix client run build	0 ERRORS (1.08S)	Clean TypeScript compilation, minified assets, compliant PWA bundle.
Backend Compilation	npm --prefix server run build	0 COMPILATION ERRORS	Zero type errors across Express routes, services, Prisma models, and socket handlers.
Database Integrity	Prisma Client DB Query	AUDITED CLEAN	100% free of legacy demo records; schema aligned for Supabase PostgreSQL.

6 IMMEDIATE NEXT STEPS & PRODUCTION ROADMAP

Cloud Storage Bucket

Connect live Supabase Storage credentials for permanent remote asset retention.

Offline Caching Cache

Expand service worker caching to support offline read-only learning material access.

Cohort Analytics

Tutor grading export to CSV and aggregate cohort progress reporting.

Report Verification & Engineering Sign-Off:

This document certifies that the Knowvia platform has achieved 100% of its foundational architecture, security hardening, and department-scoped functional requirements for the 2026 Tech Hub Internship Cohort.

NASCOM / Sydon
Lead Platform Engineering
Verified: September 15, 2026